

Tutorial 1 Memo

Experiment Description

Does the amount of therapy have an effect on the level of test anxiety students experience? Researchers were interested in this question and compared test anxiety under different lengths of therapy: 5 hours, 10 hours and 15 hours. The length of therapy was randomly assigned to each of 15 students resulting in three groups of 5 students. At the end of therapy, each subject completed an evaluation of test anxiety (the dependent variable in the study).

The three groups of students received the scores on the Test Anxiety Index (TAI) at the end of treatment shown in the table below and in the csv file called “test_anxiety.csv”. Open a new Rscript and load the data, inspect the data set and identify the column names. Compare the format of the data (long format) in the csv file to what is shown below (wide format).

Group 1	Group 2	Group 3
48	55	51
50	52	52
53	53	50
52	55	53
50	53	50

1. Identify the following based on the experiment description:

- Treatment factor: **Therapy duration**
- Factor levels: **5 hours, 10 hours, 15 hours**
- Treatments: **same as levels**
- Response: **Test Anxiety Index (TAI) scores**
- Experimental units: **Individual students**
- Number of replicates: **5 per treatment group**

2. What is the treatment structure and how has randomization been conducted?

The treatment structure is **single-factor with three levels**. Randomization is conducted by **randomly assigning students to one of the three therapy groups** without blocking.

3. Based on the treatment structure and randomisation procedure you have identified what type of design has been used here?

This is **single-factor completely randomised design**.

4. What would be the problem if we said group was the experimental unit?

If groups were considered experimental units, we would only have 1 replicate per treatment instead of 5. We would not be able to distinguish a treatment effect from the effect of experimental unit (i.e. the effect of belonging to a certain group).

5. Write down the population model equation used for analysing data from this type of experiment. Be specific.

$$y_{ij} = \mu + A_i + \varepsilon_{ij}$$

where:

- y_{ij} = response variable (TAI score) where $i = 1, 2, 3$ and $j = 1, \dots, 5$
- μ = overall mean
- A_i = treatment effect for therapy duration
- ε_{ij} = random error term and $\varepsilon_{ij} \sim N(0, \sigma^2)$

6. Calculate the following by hand and note down the R code you could use to calculate the first three:

- a. Overall meanY

$$\begin{aligned}\bar{Y}_{..} &= \frac{1}{N} \sum Y_{ij} \\ &= \frac{48 + 50 + 53 + 52 + 52 + 50 + 55 + 52 + 53 + 55 + 53 + 51 + 52 + 50 + 53 + 50}{15} \\ &= 51.8\end{aligned}$$

Rcode: `mean(data$Score)`

- b. Treatment means

For first group (5 hours):

$$\bar{Y}_{1.} = \frac{1}{r} \sum Y_{1j} = \frac{48 + 50 + 53 + 52 + 52 + 50}{5} = 50.6$$

Rcode: `mean(data$Score[data$Group == "Group 1"])`

For second group (10 hours):

$$\bar{Y}_{2.} = \frac{1}{r} \sum Y_{2j} = \frac{55 + 52 + 53 + 55 + 53}{5} = 53.6$$

Rcode: `mean(data$Score[data$Group == "Group 2"])`

For third group (15 hours):

$$\bar{Y}_{3.} = \frac{1}{r} \sum Y_{3j} = \frac{51 + 52 + 50 + 53 + 50}{5} = 51.2$$

Rcode: `mean(data$Score[data$Group == "Group 3"])`

c. Treatment effects

$$A_i = \mu_i - \mu$$

Treatment effect 1: $A_1 = 50.6 - 51.8 = -1.2$

Treatment effect 2: $A_2 = 53.6 - 51.8 = 1.8$

Treatment effect 3: $A_3 = 51.2 - 51.8 = -0.6$

d. Estimated error variance

$$\begin{aligned} s^2 &= \frac{1}{N - a} \sum_{ij} (Y_{ij} - \bar{Y}_{i.})^2 \\ &= \frac{1}{N - a} \left(\sum_j (Y_{1j} - \bar{Y}_{1.})^2 + \sum_j (Y_{2j} - \bar{Y}_{2.})^2 + \sum_j (Y_{3j} - \bar{Y}_{3.})^2 \right) \end{aligned}$$

For the first sum in the brackets:

$$\begin{aligned} \sum_j (Y_{1j} - \bar{Y}_{1.})^2 &= (48 - 50.6)^2 + (50 - 50.6)^2 + (53 - 50.6)^2 + (52 - 50.6)^2 + \\ &\quad (50 - 50.6)^2 = 15.2 \end{aligned}$$

$$\sum_j (Y_{2j} - \bar{Y}_{2.})^2 = 7.2$$

$$\sum_j (Y_{3j} - \bar{Y}_{3.})^2 = 29.2$$

$$s^2 = \frac{1}{N-a} \left(\sum_j (Y_{1j} - \bar{Y}_{1.})^2 + \sum_j (Y_{2j} - \bar{Y}_{2.})^2 + \sum_j (Y_{3j} - \bar{Y}_{3.})^2 \right) = \frac{1}{15-3} (15.2 + 7.2 + 6.8) = 2.43333$$

7. Write out the separate model equations for each treatment.

- Group 1: $y_{1j} = \mu + A_1 + \varepsilon_{1j}$
- Group 2: $y_{2j} = \mu + A_2 + \varepsilon_{2j}$
- Group 3: $y_{3j} = \mu + A_3 + \varepsilon_{3j}$

Where each $\varepsilon_{ij} \sim N(0, \sigma^2)$

8. What is the error term for observation Y_{34} ?

$$Y_{34} = 53$$

$$\bar{Y}_{3.} = 51.2$$

$$e_{34} = Y_{34} - \bar{Y}_{3.} = 53 - 51.2 = 1.8$$

9. What is the standard error of treatment means? Of differences between treatment means? Calculate without using R (you won't have R in the exam).

Treatment means:

$$SE(\bar{y}_i) = \sqrt{s^2/n} = \sqrt{2.433333/5} = 0.6976145 \approx 0.70$$

Difference in means:

$$SE(\bar{y}_i - \bar{y}_j) = \sqrt{2 * (s^2/n)} = \sqrt{2 * (2.433333/5)} = 0.98657.. \approx 0.99$$

10. What are the assumptions of this model and how would you check the validity of each of these?

1. No outliers: boxplots / visual checks
2. Normality of errors: Check using QQ-plots and boxplots before model fitting
3. Homogeneity of Variance: Visual checks and looking at sample statistics (IQR and sd)
4. Independence: Careful consideration of the design protocol and dotchart of observations before model fitting and of residuals after model fitting.

11. Plot the data per treatment as box plots with the points overlayed. Select the options that apply:
- a. The boxplots indicate that are severe outliers.
 - b. It is hard to tell if the data are normally distributed because the sample sizes are so small.
 - c. The heights of the boxplots look comparable.
12. Which assumption(s) do we need to check after model fitting as well? Why is that?

Normality and independence, the assumptions are about the model errors and not the raw response.

13. What is the ratio of largest to smallest standard deviations of the treatments and what does this indicate to you?

$$1.949359/1.303840 = 1.495...$$

This ratio is small (less than 5) which suggest that the assumption equal population variance is reasonable.

14. If the sample standard deviations are different that means that the groups do not have equal population variance. True or false? Motivate your answer.

FALSE. Just because the sample standard deviations are different it does not mean the groups do not have equal population variance. This is because of sampling variability, every time we sample from a population the values are slightly different and so the resulting sample statistics (i.e. standard deviation) will be different, but it still comes from the same population.

Question 2

A food scientist is investigating whether four different baking temperatures affect the average rise height (cm) of a particular type of bread.

- The four temperatures are: 180°C, 190°C, 200°C, and 210°C.
- For each temperature, the scientist prepares 12 loaves of bread, using the same recipe and ingredients.
- Each loaf is baked independently in the oven at the assigned temperature.
- After baking, the rise height of each loaf is measured once.
- The loaves are baked in a random order.

1. How many replicates does this experiment have?

There are 12 replicates per treatment .

2. What is the experimental unit?

The experimental unit is an individual loaf of bread.

3. Why is it important that each loaf is baked independently?

Independence ensures that the observations are not related to one another.

4. TRUE OR FALSE: Randomising the baking order helps ensure that differences in rise height are due to temperature rather than time-related effects.

5. Is this experiment balanced or unbalanced?

Balanced (there are an equal number of replicates per treatment).

6. Suppose the scientist measured the rise height of each loaf three times and recorded the average. Would this change the number of replicates? Explain.

No, this would not change the number of replicates. Replication depends on the number of independent experimental units, not the number of measurements. Multiple measurements on the same loaf are repeated measurements, not additional replicates.

7. If the loaves had been baked one temperature per day (all 180°C on Day 1, all 190°C on Day 2, etc.), what potential problem could arise?

A confounding problem could arise. Day-to-day variation such as changes in oven performance or environmental conditions would be confounded with temperature, making it difficult to attribute differences in rise height solely to the baking temperature.

8. State the null and alternative hypothesis for analysing this experiment using ANOVA.

Null hypothesis (H_0):

All baking temperatures result in the same mean rise height.

Alternative hypothesis (H_1):

At least one mean rise height differs between temperatures.

9. Write down the model equation used for analysing data from this type of experiment. Be specific.

$$y_{ij} = \mu + A_i + \varepsilon_{ij}$$

where:

- y_{ij} = response variable (Rise Height) where $i = 1, 2, 3, 4$ and $j = 1, \dots, 12$
- μ = overall mean rise height
- A_i = treatment effect for temperature
- ε_{ij} = random error term and $\varepsilon_{ij} \sim N(0, \sigma^2)$

10. Use the following code in R to simulate data and check the assumptions.

See the R script. Even though the data were simulated from a normal distribution, small samples can produce Q-Q plots that do not look perfectly straight. In ANOVA, only extreme departures from normality are a concern, and the deviations observed here are not severe.

A possible outlier is present, but it is not extreme. A sensible approach is to refit the model with and without this observation to assess the robustness of the results.

More questions

To answer these questions, use the lecture slides and notebook.

2. Explain the difference between an observational study and an experiment.

In an observational study, the researcher does not actively manipulate variables and only observes what naturally occurs. In an experiment, the researcher

deliberately manipulates one or more factors and controls how treatments are assigned to experimental units.

3. What is one key principle of experiments that allows the establishment of causal relationships?

Randomisation. Random assignment of treatments helps balance confounding variables across groups, allowing causal effects to be identified.

4. Why is it difficult to establish causal effects through observational studies?

Because of confounding. Other variables may influence both the explanatory variable and the response, making it difficult to isolate causal effects.

5. Define the following terms in terms of experimental design:

a. Response:

The outcome measured in the experiment.

b. Factor:

A variable deliberately manipulated by the researcher.

c. Treatments:

The specific levels or combinations of factor levels applied to experimental units.

d. Experimental units:

The individuals or objects to which treatments are applied.

6. What is an experiment called where treatments are combinations of factor levels?

A factorial experiment.

7. Why is having more than one response value per experimental unit problematic?

It violates the assumption of independence. Multiple measurements on the same experimental unit are likely correlated, leading to pseudoreplication.

8. What design principle can be used when experimental units are not homogenous?
How is randomization applied in this procedure?

Blocking. Experimental units are grouped into blocks of similar units, and treatments are randomly assigned within each block.

9. Suppose a researcher conducts 10 independent hypothesis tests, each at a 5% significance level. If all null hypotheses are true, calculate the probability that the researcher makes at least one Type 1 error.

$$P(\text{at least one Type I error}) = 1 - (1 - 0.05)^{10} \approx 1 - 0.5987 = 0.4013.$$